

Rational Functions and Asymptotes



Finding asymptotes and intercepts

- 1 For $f(x) = \frac{x+2}{x-3}$, find:
 - a the vertical asymptote
 - b the horizontal asymptote
 - c the x-intercept
 - d the y-intercept
 - 2 Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.
 - 3 Consider $f(x) = \frac{x^2-9}{x-3}$.
 - a Simplify f and identify the feature at $x = 3$.
 - b Does the graph have a vertical asymptote?
 - 4 Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.
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Behaviour near asymptotes

- 5 For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:
 - a $x \rightarrow 3^+$
 - b $x \rightarrow 3^-$
 - c $x \rightarrow \infty$
 - 6 Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.
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Classifying end behaviour from degree

- 7 Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:
 - a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2)
 - b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2)
 - c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2)
 - 8 Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.
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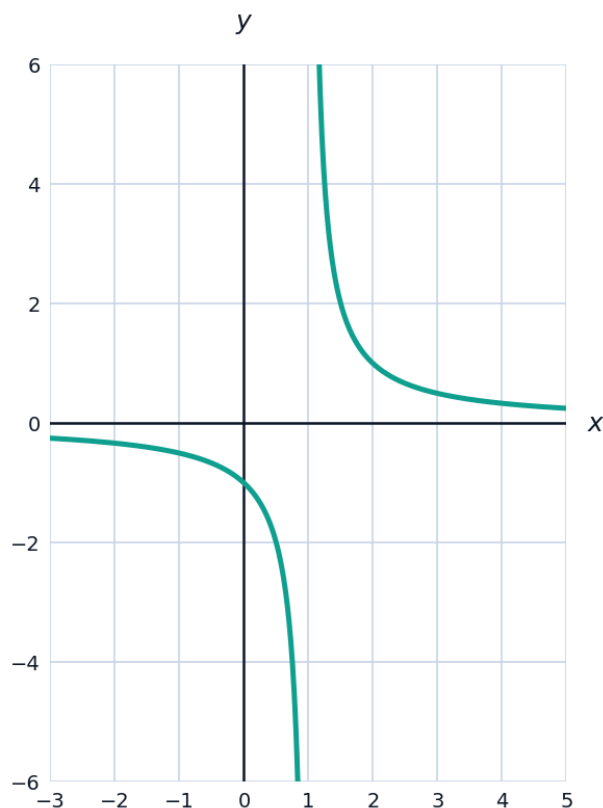
Holes versus asymptotes

9 For $f(x) = \frac{(x-2)(x+5)}{(x-2)(x-1)}$, identify the hole and the vertical asymptote.

10 Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.

Reading asymptotic behavior from a graph

11 The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \rightarrow \pm \infty$.

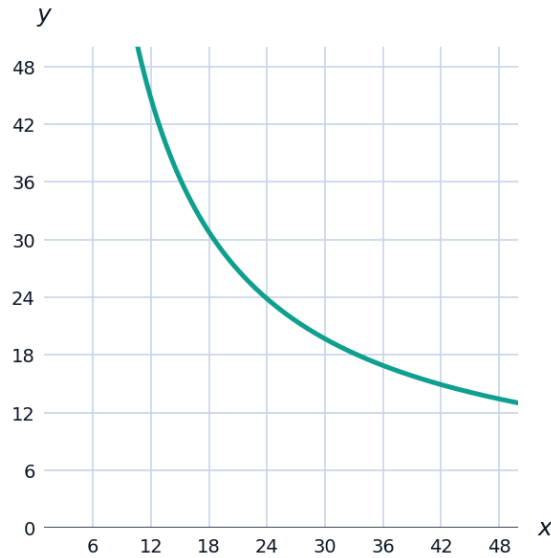


Domain restrictions and sign analysis

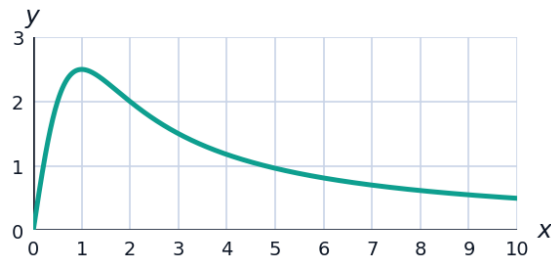
12 Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.

Asymptotes in applied contexts

- 13 The average cost per unit of producing x items is $C(x) = \frac{500 + 3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.



- 14 A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2 + 1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.



Finding a rational function's equation from its features

- 15 Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x -intercept at $x = 1$.